Supplemental Material

Data S1.

Search criteria for the meta-analysis on 30/03/2020

Ovid MEDLINE(R) ALL (1946 to March 26, 2020)

#	Search Statement	Results
1	myocardial strain.mp.	1118
2	deformation imaging.mp.	305
3	speckle tracking echocardio*.mp.	2750
4	speckle tracking stud*.mp.	56
5	speckle tracking analys*.mp.	245
6	deformation analys*.mp.	379
7	diabetes mellitus/ or diabetes mellitus, type 1/ or diabetes mellitus, type 2/	299992
8	diabetes mellitus.mp.	427259
9	3 or 4 or 5	2946
10	7 or 8	427259
11	1 and 2 and 9	10
12	1 or 2 or 6 or 9	4330
13	10 and 12	160
14	(left ventric* or LV or right ventric* or RV or left atri* or LA or right atri* or RA).mp.	881773
15	(function* or dysfunction*).mp.	4093791
16	14 and 15	191576
17	strain.mp.	430336
18	16 and 17	8130
19	13 and 18	127
20	13 or 19	160
21	Filter by journal article and human studies	121

Embase Classic+Embase (1947 to March 27, 2020)

	Search Statement	Results
1	myocardial strain.mp.	2398
2	deformation imaging.mp.	683
3	speckle tracking echocardio*.mp.	8331
4	speckle tracking stud*.mp.	171
5	speckle tracking analys*.mp.	860
6	deformation analys*.mp.	689
7	diabetes mellitus/ or diabetes mellitus, type 1/ or diabetes mellitus, type 2/	659383
8	diabetes mellitus.mp.	954813
9	3 or 4 or 5	8984
10	7 or 8	954813
11	1 and 2 and 9	22
12	1 or 2 or 6 or 9	11581
13	10 and 12	794
14	(left ventric* or LV or right ventric* or RV or left atri* or LA or right atri* or RA).mp.	1443344
15	(function* or dysfunction*).mp.	5908112
16	14 and 15	350926
17	strain.mp.	842307
18	16 and 17	20484
19	13 and 18	574
20	13 or 19	794
21	Filter by journal articles and human studies	330

Cochrane central register of controlled trials (1991 to February 2020)

#	Search Statement	Results
1	myocardial strain.mp.	101
2	deformation imaging.mp.	16
3	speckle tracking echocardio*.mp.	300
4	speckle tracking stud*.mp.	3
5	speckle tracking analys*.mp.	33
6	deformation analys*.mp.	12
7	diabetes mellitus/ or diabetes mellitus, type 1/ or diabetes mellitus, type 2/	25548
8	diabetes mellitus.mp.	63955
9	3 or 4 or 5	324
10	7 or 8	63955
11	1 and 2 and 9	0
12	1 or 2 or 6 or 9	417
13	10 and 12	26
14	(left ventric* or LV or right ventric* or RV or left atri* or LA or right atri* or RA).mp.	55341
15	(function* or dysfunction*).mp.	278968
16	14 and 15	20338
17	strain.mp.	8265
18	16 and 17	803
19	13 and 18	19
20	13 or 19	26
21	Filter by journal articles and human studies	10

Scopus (March 30, 2020): 40 studies

(TITLE-ABS
KEY (myocardial AND strain OR deformation AND imaging OR speckle AND tracking AND stud* OR speckle AND tracking AND analys* OR deformation AND analys*))
AND (TITLE-ABS-
KEY (diabetes AND mellitus OR diabetes AND mellitus AND type 1 OR diabetes AN D mellitus AND type 2)) AND (LIMIT
TO (LANGUAGE , "English")) AND (LIMIT-
TO (DOCTYPE , "ar")) AND (LIMIT-TO (SUBJAREA , "MEDI")) AND (LIMIT-
TO (EXACTKEYWORD , "Human"))

Web of Science (March 30, 2020): 290 studies

(myocardial AND strain OR deformation AND imaging OR speckle AND tracking AN D stud* OR speckle AND tracking AND analys* OR deformation AND analys*) AN D (diabetes AND mellitus OR diabetes AND mellitus AND type 1 OR diabetes AND mellitus AND type 2) Refined by: LANGUAGES: (ENGLISH) AND DOCUMENT TYPES: (ARTICLE) AND WEB OF SCIENCE CATEGORIES: (CARDIAC CARDIOVASCULAR SYSTEMS OR ENDOCRINOLOGY METABOLISM)

Table S1. Definition of study groups of included studies.

First Author	Year	DM group inclusion criteria	DM group exclusion criteria	Control group definition
Zhang	2013	Type 2 DM with LVEFs > 55%	Arrhythmias, ischemic heart disease (documented myocardial infarction, history of revascularization procedures, and positive findings on coronary angiography or computed tomography), other structural heart diseases, albumin/creatinine ratios > 30 mg/mg, uncontrolled hypertension (systolic blood pressure > 140 mm Hg and/or diastolic blood pressure > 90 mm Hg at rest), and one with other serious complications of DM (diabetic ketoacidosis or nonketotic hyperosmolar coma), inadequate echocardiographic image quality	Age-matched and gender-matched controls if they met the following criteria: no evidence of pre-existing cardiac disease on transthoracic echocardiography and other examinations, no clinical history of chronic diseases or chronic medications, and normal results on 12-lead electrocardiography. However, subjects with hypertension but well-controlled blood pressure (i.e., systolic blood pressure < 140 mm Hg and diastolic blood pressure < 90 mm Hg on three separate measurements during the past month) were not excluded.
Wang-1	2015	Outpatients or inpatients diagnosed with type 2 diabetes according to the 2010 guidelines of the American Diabetes Association, and with left ventricular ejection fraction (LVEF) > 55%	Presence of coronary artery stenosis, atrial fibrillation, or poor acoustic window and image quality	Age- and gender-matched controls were recruited from the medical examination centre in our hospital based on the following criteria: no history of diabetes, hypertension, or coronary heart disease and no evidence of pre-existing cardiac diseases on conventional electrocardiogram (ECG), echocardiography, or laboratory examination
Tadic	2015	Normotensive patients ((blood pressure < 140/90 mm Hg measured on several separate occasions) untreated for type 2 diabetes	Symptoms or signs of cardiovascular disease (arterial hypertension, heart failure, myocardial infarction, significant valvular disease, atrial fibrillation, congenital heart disease), obesity (BMI ≥ 30 kg/m2), asthma, chronic obstructive lung disease, neoplastic disease, cirrhosis of the liver, or kidney failure	Controls of similar age and sex distributions
Wang-2	2015	Patients diagnosed with type 2 diabetes [left ventricular ejection fraction (LVEF) ≥55%]	History of coronary artery stenosis diagnosed by coronary angiography, arrhythmia, or poor acoustic windows and imaging qualities	Age- and sex-matched controls with BMIs of 18.5–24.5 kg/m2 who showed an absence of diabetes, hypertension, and coronary heart disease and in whom there was no evidence of pre-existing cardiac disease on conventional electrocardiography (ECG), transthoracic echocardiography (TTE), and laboratory examinations

Enomoto	2016	Hospitalized patients with type 2 DM requiring diabetes education	Patients with coronary artery disease or LV ejection fraction (LVEF) <50%, other than sinus rhythm, significant valvular disease, and inadequate echocardiographic image quality for analysis	Age-matched healthy subjects
Wang	2017	Type 2 diabetic patients aged 60 years or over	Arterial hypertension, coronary lesions, severe arrhythmia, and other heart disease with known causes	Sex-matched healthy volunteers aged over 60 years, without diabetes, hypertension, or pre-existing cardiac disease on conventional examinations
Luo	2018	T2DM complicated with microangiopathy. Microangiopathies were as follows: (1) diabetic retinopathy, (2) diabetic nephropathy, (3) diabetic neuropathy	Acute complications, severe hypoglycaemia during treatment, hypertension, coronary heart disease, and arrhythmia	Healthy volunteers not have a history of diabetes mellitus, hypertension and coronary heart disease
Ringle	2018	Patients aged older than 18 years with isolated type 1 diabetes	Recent diagnosis of diabetes (<1year), documented cardiac disease, diabetic nephropathy, cardiovascular risk factors (hypertension, hypercholesterolemia, active smoking, obesity, age over 60 years)	Age- and gender-matched healthy subjects that met the following criteria: no cardiovascular risk factors, no personal history of heart disease, no clinical history of chronic disease or chronic medication and normal transthoracic echocardiography
Wang-1	2018	(1) diagnosis of type 2 diabetes based on the 2014 guidelines of the American Diabetes Association10; and (2) a left ventricular ejection fraction (LVEF) of 55% or higher	(1) patients who had a history of hypertension or coronary artery stenosis diagnosed by coronary angiography and computed tomography; (2) patients with arrhythmia; (3) patients who had known causes of chronic liver disease (i.e., alcoholic or drug-induced liver disease and autoimmune or viral hepatitis); and (4) patients with a poor acoustic window and fuzzy 3D images	Age- and sex-matched healthy Volunteer, absence of diabetes, hypertension, and coronary heart disease and no evidence of pre-existing cardiac or hepatic disease in conventional radiologic, US, or laboratory tests
Wang-2	2018	Type 2 diabetic patients	Possibility of hypertension, coronary lesion, severe arrhythmia, and other heart diseases with known causes	Age-and sex-matched healthy volunteers from our medical examination centre were recruited as the control group. All of them showed the absence of diabetes, hypertension, or other pre-existing cardiac diseases using conventional examinations, and they also had serum concentrations of cholesterol and triglycerides within the normal range
Wang-1	2019	Diagnostic criteria of T2DM were in line with the guidelines	Presence of hypertension, coronary lesion, severe arrhythmia, and other heart diseases with known causes	Age- and gender-matched healthy physical examinees in our hospital were recruited as the

		promulgated by the American Diabetes Association (ADA) in 2016		control group. None of the controls had obviously positive signs of T2DM, hypertension, or other pre-existing cardiac diseases after routine examinations. Moreover, they all had SUA at normal levels.
Wang-2	2019	The diagnostic criteria of T2DM complied with the 2016 guidelines of the American Diabetes Association (ADA).	Hypertension, severe arrhythmia and valve lesion, as well as other cardiac disease with known causes	Age-and gender-matched healthy individuals who received regular check-ups in our hospital were recruited as the control subjects. They all had no history or positive signs of T2DM, hypertension, coronary disease, and other pre-existing heart diseases detected by routine examinations

Table S2. Haemodynamic parameters in included articles.

First Author	Year	Rest SBP Mean ± SD (DM)	Rest SBP Mean ± SD (Control)	Rest DBP Mean ± SD (DM)	Rest DBP Mean ± SD (Control)	Rest HR Mean ± SD (DM)	Rest HR Mean ± SD (Control)
Zhang	2013	128±8	129±12	76±15	77±7	76±15	74±13
Wang-1	2015	127±9.5	124±9.2	78±0.4	77±5.6	79±11	76±10
Tadic	2015	128±11	124±13	77±7	75±8	71±8	73±7
Wang-2	2015	127±10.2	126±8.4	77±7.3	78±4.5	78±12	75±11
Enomoto	2016	119.8±21.4	122.9±17.2	70.3±13.6	73.2±13.4	66.1±11.9	65.5±12.7
Wang	2017	128±7	123±9.2	78±5.3	75±6.4	77±12	74±13
Luo	2018	128.4±8.9	107.2±8.8	75.7±5.9	73.3±6	-	-
Ringle	2018	121±12	122±9	74±9	74±6	-	-
Wang-1	2018	127±9.8	126±6.8	78±5.8	76±6.4	77±12	75±10
Wang-2	2018	127.8±8.8	125.7±10	77.3±6.4	77.6±7.7	77±11	75±13
Wang-1	2019	128.7±5.7	126.5±6.9	75.6±7.3	75±6	76±10	77±9
Wang-2	2019	125±8.48	125±7.26	76±5.37	75±7.19	78±10	75±10

Table S3. Meta-regression results in diabetes mellitus group.

Variable		DM type (II vs I)	Vendor (Toshiba vs G.E)	Vendor (Philips vs G.E)	Age, per 1 year	%female, per 1%	SBP, per 1 mmHg	DBP, per 1 mmHg	HR, per 1 pm	%HTN, per 1%	BMI, per 1 kg/m2	HbA1C, per 1%
3D LV GLS	**N**	12	12	12	12	12	12	12	10	12	12	7
	β [95% CI]	-0.94 [-4.6, 2.7]	**-5.8 [-7, -4.6]**	-0.3 [-1.4, 0.7]	0 [-0.1, 0.1]	**0.1 [0, 0.2]**	0.3 [0, 0.6]	**0.5 [0.2, 0.9]**	**0.3 [0.1, 0.5]**	**-0.1 [-0.1, 0]**	0.7 [-0.2, 1.6]	-0.7 [-1.5, 0]
	***P*-value**	0.62	**<0.001**	0.46	0.97	**0.04**	0.07	**0.002**	**0.007**	**0.001**	0.1	0.06
3D LV GCS	**N**		11	11	11	11	11	11	10	11	11	7
	β [95% CI]		**8.8 [6.7, 10.9]**	**4 [2.3, 5.7]**	**-0.3 [-0.6, 0]**	**-0.4 [-0.6, -0.3]**	**-1 [-1.5, -0.4]**	**-1.3 [-1.8, -0.8]**	**-0.7 [-0.9, -0.4]**	0.1 [0, 0.1]	-0.7 [-2.1, 0.8]	**1.8 [0.6, 3.1]**
	***P*-value**		**<0.0001**	**<0.0001**	**0.03**	**<0.0001**	**0.0005**	**<0.0001**	**<0.0001**	0.2	0.4	**0.003**
3D LV GRS	**N**		10		10	10	10	10	10	10	10	6
	β [95% CI]		**-14.7 [-18, -11.6]**		**1 [0.2, 1.7]**	**1.1 [0.7, 1.5]**	**2.2 [1, 3.5]**	**2.8 [1.6, 3.9]**	**1.7 [1.3, 2.1]**	**-0.2 [-0.4, 0]**	0.8 [-3.7, 5.2]	**-4.7 [-7.7, -1.7]**
	***P*-value**		**<0.0001**		**0.01**	**<0.0001**	**0.0004**	**<0.0001**	**<0.0001**	**0.04**	0.7	**0.002**
3D LV GAS	**N**		10		10	10	10	10	10	10	10	6
	β [95% CI]		**8.4 [6.3, 10.5]**		-0.2 [-0.4, 0.1]	**-0.4 [-0.5, -0.2]**	**-0.9 [-1.2, -0.7]**	**-1 [-1.3, -0.6]**	**-0.5 [-0.8, -0.2]**	0.1 [0, 0.1]	-1 [-2.1, 0.1]	**1.5 [0.2, 2.7]**
	***P*-value**		**<0.0001**		0.24	**<0.0001**	**<0.0001**	**<0.0001**	**0.001**	0.09	0.07	**0.02**

Table S4. Meta-regression results in control group.

Variable		DM type (II vs I)	Vendor (Toshiba vs G.E)	Vendor (Philips vs G.E)	Age, per 1 year	%female, per 1%	SBP, per 1 mmHg	DBP, per 1 mmHg	HR, per 1 pm	%HTN, per 1%	BMI, per 1 kg/m2	HbA1C, per 1%
3D LV GLS	**N**	12	12	12	12	12	12	12	10	12	12	4
	β [95% CI]	0 [-3, 3]	**-2.9 [-5.7, -0.1]**	0.8 [-1.7, 3.2]	0 [-0.1, 0.1]	0 [-0.1, 0.1]	-0.1 [-0.2, 0.1]	0.3 [-0.2, 0.8]	**0.3 [0, 0.4]**	0 [-0.1, 0.1]	0.3 [-0.6, 1.2]	0 [-4, 4]
	***P*-value**	0.9	**0.04**	0.53	0.66	0.53	0.4	0.2	**0.02**	0.9	0.5	0.9
3D LV GCS	**N**		11	11	11	11	11	11	10	11	11	4
	β [95% CI]		**11.8 [9.5, 14.1]**	**5.5 [3.8, 7.2]**	**-0.4 [-0.7, -0.2]**	**-0.4 [-0.7, -0.1]**	**-0.4 [-0.6, -0.2]**	**-1.8 [-2.7, -0.9]**	**-1.3 [-1.5, -1]**	0 [-0.2, 0.1]	-2.2 [-4.1, -0.3]	**-1.9 [-3.4, 0.4]**
	***P*-value**		**<0.0001**	**<0.0001**	**0.0005**	**0.005**	**0.0008**	**0.0001**	**<0.0001**	0.5	0.02	**0.01**
3D LV GRS	**N**		10		10	10	10	10	10	10	10	4
	β [95% CI]		**-13.5 [-20.2, -6.9]**		**0.7 [0.2, 1.2]**	1.7 [-0.8, 4.2]	**2.3 [0.7, 4.1]**	**3.4 [1.4, 5.4]**	**1.9 [1, 2.8]**	0.2 [-0.1, 0.4]	3.9 [0, 7.8]	**11.6 [2.4, 20.8]**
	***P*-value**		**<0.0001**		**0.006**	0.18	**0.006**	**0.001**	**<0.0001**	0.2	0.05	**0.01**
3D LV GAS	**N**		10		10	10	10	10	10	10	10	4
	β [95% CI]		**13 [9.7, 16.3]**		**-0.4 [-0.6, -0.1]**	-0.8 [-2, 0.4]	-0.6 [-1.7, 0.4]	**-1.4 [-2.6, -0.2]**	**-1.2 [-1.5, -0.9]**	0 [-0.1, 0.1]	**-2.6 [-4.3, -0.9]**	-0.4 [-2.5, 1.7]
	***P*-value**		**<0.0001**		**0.007**	0.2	0.2	**0.02**	**<0.0001**	0.8	**0.003**	0.7

Table S5. Meta-regression results in diabetes mellitus group in subgroup analysis of studies using the most popular vendor.

Variable		Age, per 1 year	%female, per 1%	SBP, per 1 mmHg	DBP, per 1 mmHg	HR, per 1 pm	%HTN, per 1%	BMI, per 1 kg/m2	HbA1C, per 1%
3D LV GLS	**N**	9	9	9	9	9	9	9	5
	β [95% CI]	0. [-0.1, 0]	0.1 [0, 0.3]	-0.2 [-0.5, 0.2]	0 [-0.4, 0.5]	-0.1 [-0.3, 0]	**-0.02 [-0.04, 0]**	0.2 [-0.2, 0.5]	-0.4 [-1, 0.2]
	***P*-value**	0.4	0.1	0.4	0.9	0.2	**0.04**	0.3	0.2
3D LV GCS	**N**	8	8	8	8	8	8	8	4
	β [95% CI]	**0.1 [0, 0.3]**	**0.2 [0.1, 0.3]**	-0.2 [-0.6, 0.2]	0.1 [-0.3, -0.6]	0.3 [-0.1, 0.7]	**-0.02 [-0.04, -0.01]**	-0.1 [-0.8, 0.6]	**-0.5 [-0.9, -0.1]**
	***P*-value**	**0.01**	**0.0006**	0.3	0.6	0.2	**0.0009**	0.8	**0.007**
3D LV GRS	**N**	8	8	8	8	8	8	8	4
	β [95% CI]	0.2 [-0.1, 0.6]	0.4 [0, 0.8]	-0.6 [-1.5, 0.4]	0.3 [-0.7, 1.4]	0.4 [-0.5, 1.4]	-0.04 [-0.1, 0]	-0.4 [-1.8, 1]	-1 [-2.4, 0.3]
	***P*-value**	0.2	0.05	0.2	0.5	0.4	0.08	0.5	0.1
3D LV GAS	**N**	8	8	8	8	8	8	8	4
	β [95% CI]	0.1 [-0.1, 0.2]	0.2 [0, 0.3]	**-0.4 [-0.8, 0]**	0 [0.6, 0.5]	0.2 [-0.3, 0.7]	-0.02[-0.04, 0]	-0.2 [-0.9, 0.5]	-0.6 [-1.1, 0]
	***P*-value**	0.4	0.08	**0.03**	0.9	0.4	0.06	0.5	0.05

Table S6. Meta-regression results in control group in subgroup analysis of studies using the most popular vendor.

Variable		Age, per 1 year	%female, per 1%	SBP, per 1 mmHg	DBP, per 1 mmHg	HR, per 1 pm	%HTN, per 1%	BMI, per 1 kg/m2	HbA1C, per 1%
3D LV GLS	N	8	8	8	8	8	8	8	3
	β [95% CI]	0 [-0.1, 0.1]	0 [-0.5, 0.5]	0.1 [-0.3, 0.6]	0.5 [-0.1, 1]	**-0.2 [-1, 0.5]**	0 [0, 0]	-0.5 [-1.3, 0.2]	-2.9 [-10.1, 4.3]
	***P*-value**	0.8	0.9	0.6	0.08	**0.5**	0.9	0.2	0.4
3D LV GCS	N	8	8	8	8	8	8	8	3
	β [95% CI]	0 [-0.1, 0.1]	0.1 [-0.1, 0.4]	0 [-0.2, 0.3]	0.2 [-0.1, 0.5]	-0.3 [-0.6, 0]	0 [-0.03, 0.03]	-0.2 [-0.6, 0.2]	-0.9 [-3.2, 1.5]
	***P*-value**	0.9	0.3	0.8	0.3	0.1	0.9	0.4	0.5
3D LV GRS	N	8	8	8	8	8	8	8	3
	β [95% CI]	**-0.4 [-0.8, -0.1]**	1 [0.2, 2]	0.7 [-0.2, 1.6]	0.7 [-0.6, 2]	**-1.6 [-2.5, -0.6]**	**0.1 [0, 0.2]**	-0.1 [-2.1, 1.9]	2.4 [-5.5, 10.3]
	***P*-value**	**0.03**	0.05	0.1	0.3	**0.001**	**0.04**	0.9	0.6
3D LV GAS	N	8	8	8	8	8	8	8	3
	β [95% CI]	**-0.2 [-0.4, -0.1]**	0.3 [-0.2, 0.7]	**0.3 [0.1, 0.7]**	0.4 [-0.2, 1]	-0.5 [-1.2, -0.1]	0.05 [0.02, 0.08]	-0.3 [-1.2, 0.6]	-2.2 [-6, 1.6]
	***P*-value**	**0.002**	0.3	**0.02**	0.2	0.09	0.002	0.5	0.3

Table S7. Observer variability and strain reproducibility for included studies.

Study	N	Intra-observer variability				Inter-observer variability			
		GLS	GCS	GRS	GAS	GLS	GCS	GRS	GAS
Zhang[10] 2013	20	0.940	0.849	0.902	0.913	0.834	0.838	0.829	0.822
Wang-1[26] 2015	18	0.967	0.923	0.993	0.979	0.966	0.906	0.988	0.963
Tadic[12] 2015	-	0.95	0.87	0.7	0.89	0.91	0.83	0.65	0.86
Wang-2[11] 2015	15	0.951	0.945	0.986	0.977	0.964	0.956	0.992	0.988
Enomoto[13] 2016	10	#							
Wang[14] 2017	12	0.975	0.988	0.982	0.986	0.962	0.929	0.979	0.967
Luo[15] 2018	16	†							
Ringle[16] 2018	20	0.967	-	-	-	0.893	-	-	-
Wang-1[18] 2018	18	0.913	0.903	0.904	0.930	0.856	0.872	0.851	0.879
Wang-2[17] 2018	18	0.956	0.969	0.983	0.980	0.929	0.916	0.967	0.941
Wang-1[20] 2019	12	0.974	0.968	0.982	0.969	0.932	0.944	0.963	0.930
Wang-2[19] 2019	12	0.951	0.965	0.950	0.953	0.937	0.930	0.934	0.931

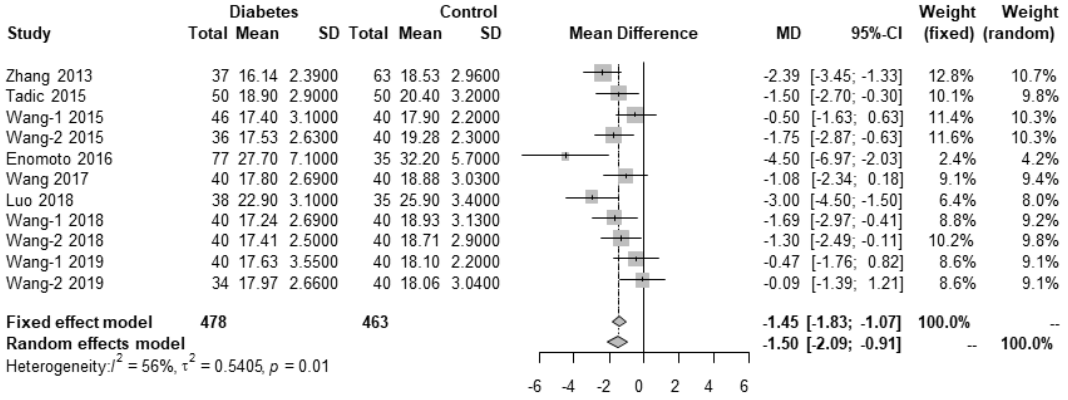
N – Number of patients assessed for variability in each study, GLS – Global Longtitudinal Strain, GCS – Global Circumferential Strain, GRS – Global Radial Strain, GAS – Global Area Strain. # - Only percent variability reported, † - Graphically represented without correlation coefficients,

Table S8. Leave-One-Out Analysis.

Study Omission	Effect	LLCI 95%	ULCI 95%	I^2
Omitting Enomoto 2016	-2.15	-2.75	-1.54	69%
Omitting Wang-2 2015	-2.16	-2.79	-1.52	73%
Omitting Wang-2 2019	-2.45	-3.15	-1.74	78%
Omitting Tadic 2015	-2.43	-3.15	-1.7	78%
Omitting Ringle 2018	-2.42	-3.15	-1.69	79%
Omitting Luo 2018	-2.23	-2.94	-1.53	79%
Omitting Wang-1 2019	-2.43	-3.14	-1.71	79%
Omitting Wang 2017	-2.43	-3.14	-1.72	79%
Omitting Zhang 2013	-2.29	-3.03	-1.56	80%
Omitting Wang-2 2018	-2.37	-3.11	-1.63	80%
Omitting Wang-1 2015	-2.36	-3.10	-1.62	80%
Omitting Wang-1 2018	-2.32	-3.06	-1.58	80%

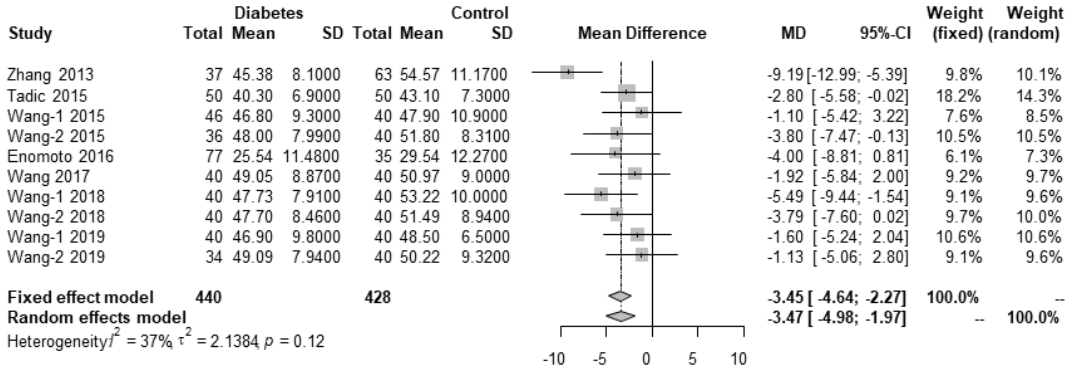
LLCI – Lower Limit Confidence Interval, ULCI – Upper Limit Confidence Interval, I^2 – Heterogeneity statistic

Figure S1. Forest plots for 3D LV GCS in diabetic and healthy controls and MD of 3D LV GCS.



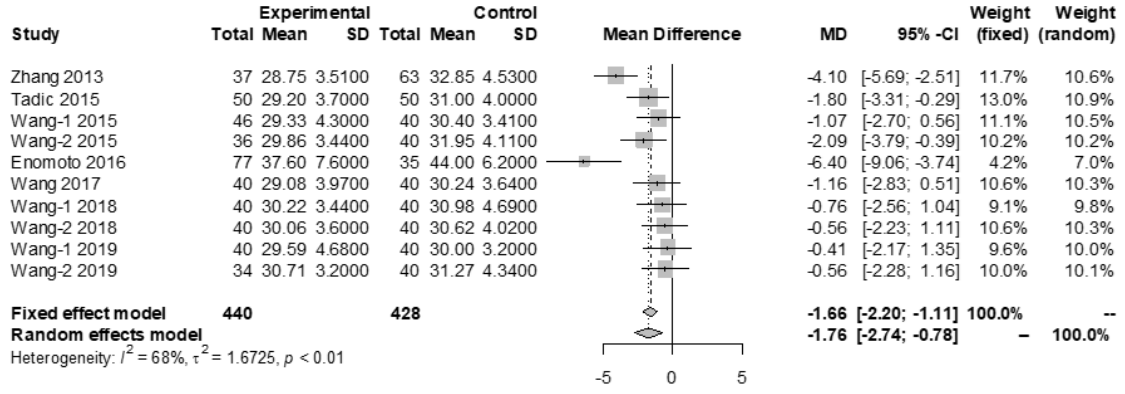
3D -three dimensional, GCS -global circumferential strain, LV - left ventricular, MD - mean difference.

Figure S2. Forest plots for 3D LV GRS in diabetic and healthy controls and MD of LV GRS.



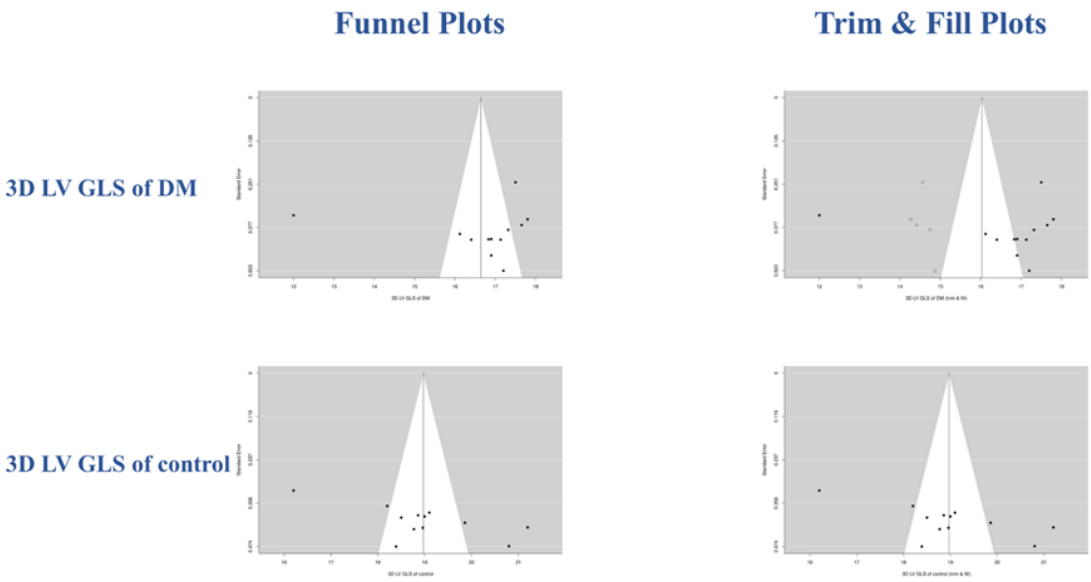
3D - three dimensional, GRS - global radial strain, LV -left ventricular, MD - mean difference.

Figure S3. Forest plots for 3D LV GAS in diabetic and healthy controls and MD of LV GAS.



3D - three dimensional, GAS - global area strain, LV - left ventricular, MD - mean difference.

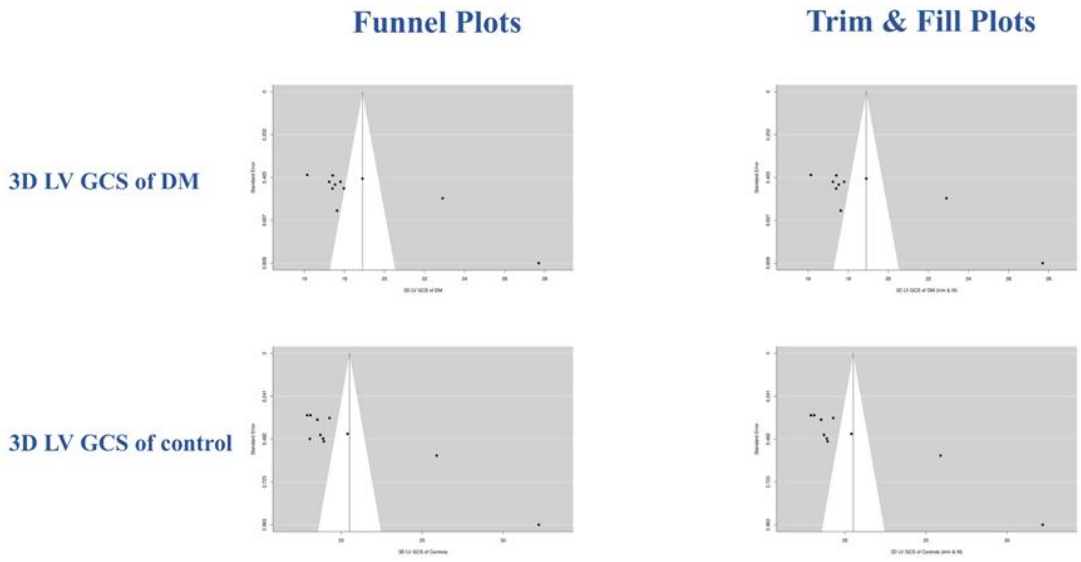
Figure S4. Funnel plots with Trim and Fill plots for 3D LV GLS in diabetic and healthy controls.



For assessment of publication bias, funnel plots of each strain parameter are shown with their trim and fill plots.

3D - three dimensional, GLS -global longitudinal strain, LV - left ventricular, MD - mean difference.

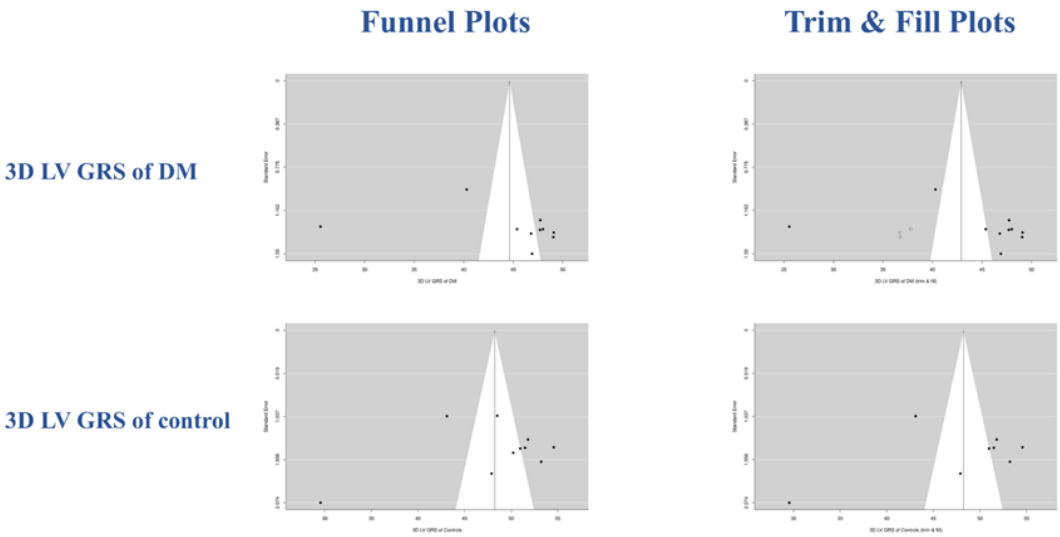
Figure S5. Funnel plots with Trim and Fill plots for 3D LV GCS in diabetic and healthy controls.



For assessment of publication bias, funnel plots of each strain parameter are shown with their trim and fill plots.

3D - three dimensional, GCS - global circumferential strain, LV - left ventricular, MD - mean difference.

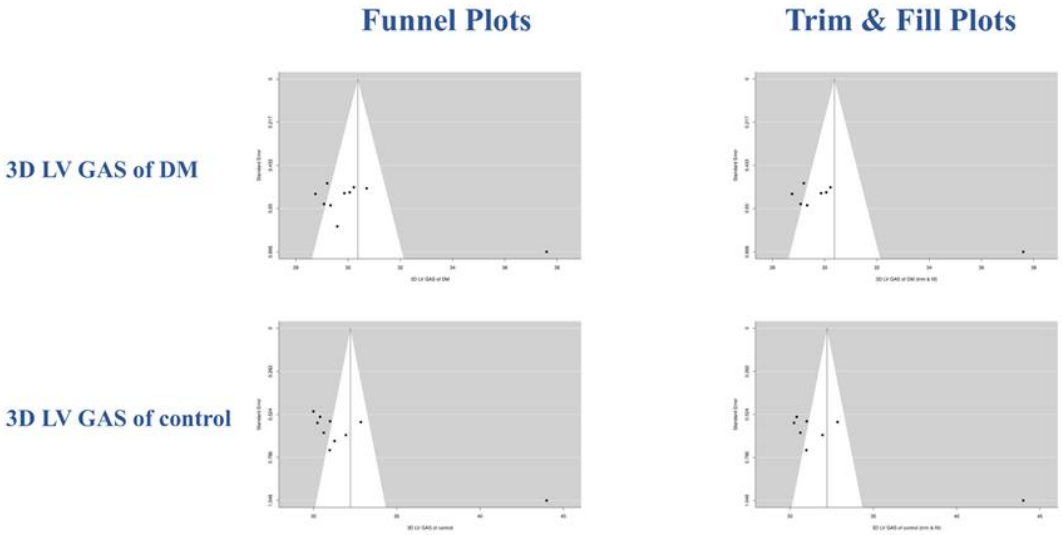
Figure S6. Funnel plots with Trim and Fill plots for 3D LV GRS in diabetic and healthy controls.



For assessment of publication bias, funnel plots of each strain parameter are shown with their trim and fill plots.

3D - three dimensional; GRS - global radial strain, LV - left ventricular, MD - mean difference.

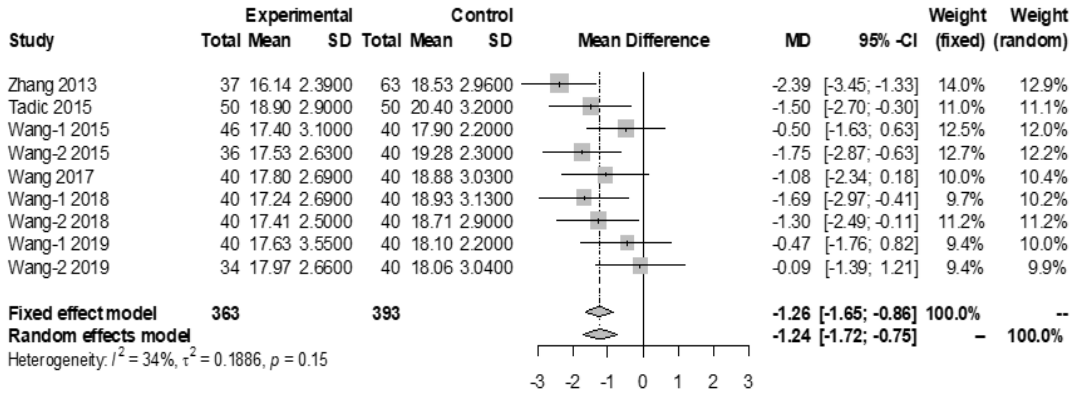
Figure S7. Funnel plots with Trim and Fill plots for 3D LV GAS in diabetic and healthy controls.



For assessment of publication bias, funnel plots of each strain parameter are shown with their trim and fill plots.

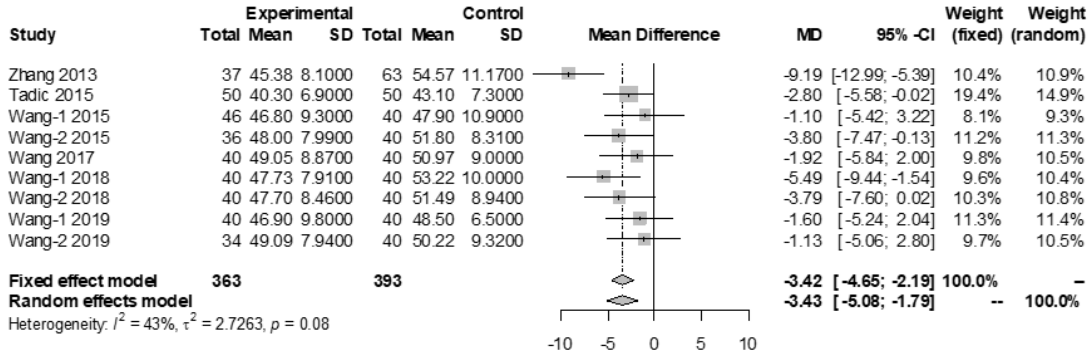
3D - three dimensional; GAS - global area strain, LV - left ventricular, MD - mean difference.

Figure S8. Forest plots of subgroup analysis of mean 3D LV GCS and MD in diabetes mellitus and control groups in studies using the most popular vendor.



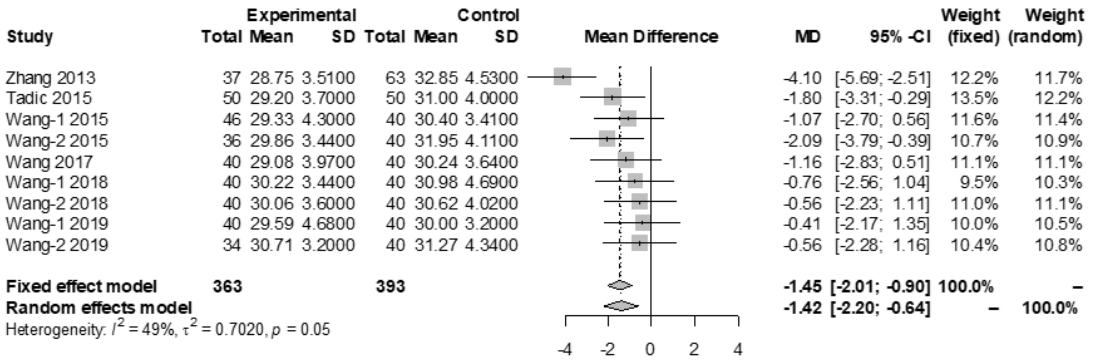
3D - three dimensional, GRS - global radial strain, LV - left ventricular.

Figure S9. Forest plots of subgroup analysis of mean 3D LV GRS and MD in diabetes mellitus and control groups in studies using the most popular vendor.



3D - three dimensional, GRS - global radial strain, LV - left ventricular.

Figure S10. Forest plots of subgroup analysis of mean 3D LV GAS and MD in diabetes mellitus and control groups in studies using the most popular vendor.



3D - three dimensional, GRS - global radial strain, LV - left ventricular.